Tianhao Li

tianhao.li | tianhao.li2@duke.edu | LinkedIn | Google Scholar | Orcid | Github

Education Degrees

Duke University **August 2024 - May 2026 (Expected)**

Master of Science Candidate in Medical Physics

Research Focus: Trustworthy AI; Large Language Model; Medical Image Analysis (Base at Duke Kunshan for the first year)

Supervisors: Prof. Fang-Fang Yin and Prof. Zhenyu Yang

North China University of Technology **September 2020 - June 2024**

Bachelor of Engineering in Information Security *GPA: 3.89/4 Grade: 90.88/100 transcript*

Core Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

Honors: Outstanding Dissertation Award (Rank 1/101, Top 0.99%), Elite Student Scholarship (Rank 1/329, Top 0.30%)

Academia Research Experiences

University of Wisconsin, Madison **October 2024 – Present**

Research Intern @ Safe And secure Foundation mOdel systems Lab (SaFoLab) *Wisconsin, United States*

- **Duties**: Led a research team to advance the AgentFuzzer project, proposing a unified agent schema descriptor, AgentSchemaDescriptor. Developed the first systematic fuzz testing framework for LLM agents, addressing the challenge of improving generalizability in agent safety alignment testing. (To be submitted to **ICML**'25)
- **Mentor**: Prof. Dr. Chaowei Xiao

Zhejiang University **July 2024 – October 2024**

Visiting Researcher @ Knowledge Engine Lab *Zhejiang, China*

- **Duties**: Led a 15-member research team on the SciSafeEval project, proposing the first safety benchmark for LLMs in scientific tasks. Identified misuse risks in SOTA LLMs (including GPT-4o, LLaMa3.2, QWen2.5) across biology, chemistry, medicine, and physics, and proposed mitigation strategies using Chain-of-Thought (CoT) and few-shot learning techniques.
- **Impacts**: arXiv:2410.03769 (**ICLR**'25 Subm.) | Huggingface:Tianhao0x01/SciSafeEval | Github:DavidLee528/SciSafeEval
- **Mentors**: Prof. Dr. Qiang Zhang, Prof. Dr. Huajun Chen

High-Tech Innovation Institute **July 2022 – July 2023**

Research Assistant @ VR & AR Technology Innovation Center *Beijing, China*

- **Duties**: Led an 8-member R&D team developing a gait-based Intelligence, Surveillance, and Reconnaissance (ISR) system. Implemented a gait recognition SaaS prototype system and proposed security features including authentication, tracking, and early warning for physical attacks. Directed technical architecture design, technical reports, and research paper writing.
- **Impacts**: **Applied 100K+ CNY funding**, published 7 SCI/EI papers, won 4 national professional contests, delivered 3 oral presentations and 1 poster at international conferences.
- **Mentors**: Prof. Dr. Lijun Wang

Industry Research Experiences

TOPSEC **March 2024 – September 2024**

Security Researcher @ Department of Innovative Technology Research *Beijing, China*

- **Duties**: **(i)** Contributed to the development of a pioneering prototype system for machine learning model security assessment, focusing on the design and implementation of a robust backend evaluation module. Successfully debugged and mitigated adversarial and poisoning attacks, enhancing the system's resilience and accuracy.; **(ii)** Contribute to a phishing email detection project, overseeing dataset collection and enhancement initiatives. Successfully applied Large Language Models (LLMs) and Supervised Fine Tuning (SFT) to develop a robust detection system.
- **Supervisor**: GM of Innovation Research Dr. Wei Wang

NSFOCUS **September 2023 – March 2024**

Security Researcher @ Dubhe Lab, Innovation Institute *Beijing, China*

- **Duties**: **(i)** Design core algorithms and develop a prototype for NSFOCUS LLM Security Assessment System (LSAS); **(ii)** Enhance the security & privacy performance of NSFGPT and other private LLMs through offensive testing, i.e. red teaming; **(iii)** Conduct literature reviews encompassing research papers and Gartner's consulting reports on trustworthy AI, enriching the threat intelligence database; **(iv)** Write white paper on secure LLM.
- **Impacts**: The release of LSAS, SecLLM NSFGPT, and white paper Enhancing Network Security with SecLLM.
- **Supervisor**: Principal Security Researcher Dr. Xingkai Wang, CIO & GM of Innovation Research Dr. Wenmao Liu

Selected Publications

- Li, T., Lu, J., Chu, C., Zeng, T., Zheng, Y., Li, M., Huang, H., Wu, B., Liu, Z., Ma, K., Yuan, X., Wang, X., Ding, K., Chen, H., & Zhang, Q.*. SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks. *International Conference on Learning Representations, 2025 (ICLR Submission)* [PDF]
- Li, T., Ma, W., Zheng, Y., Fan, X., Yang, G., Li, Y., Wang, L., & Li, Z.*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology. *PeerJ Computer Science, 2024 (SCI, JCR-Q1, Accepted)* [Poster]
- Ma, W., Li, Y.*, Li, T., Yang, H., Li, Z., Wang, L., & Xuan, J.. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *Neurocomputing, 2024 (SCI, JCR-Q1, Peer Review)*
- Ma, W., Zheng, Y., Li, T., Li, Z., Li, Y., & Wang, L.*. A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications. *PeerJ Computer Science, 2024 (SCI, JCR-Q1)* [PDF]
- Li, T., Zheng, Y., Ma, W., Wang, G., Li, Z., & Wang, L.*. Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Zheng, Y., Li, T., Ma, W., Zheng, J., Li, Z.*, & Wang, L.. Unveiling Privacy Challenges: Big Data-Driven Digital Twins in Smart City Applications. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Zheng, Y., Li, T., Ma, W., Li, Z., Wang, L., & Li, Y.*. Automated Pricing and Replenishment Decision for Vegetable Products Based on Hybrid Machine Learning Models. *Electronics, Communications and Networks, 2024* [PDF][Oral]
- Ma, W., Li, Z., Li, T., Zheng, Y., & Wang, L.. Application of Virtual Reality Technology in the Diagnosis and Treatment of Psychological Disorders: An Electroencephalography (EEG)-Based Approach. *SID Symposium Digest of Technical Papers, 2024 (ICDT)* [Oral]
- Li, T., Zheng, Y., Zhang, H., Ma, W., & Li, Y.*. Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. *Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, 2023 (BDIoT)* [PDF][Oral][Slide]

Honors and Awards

- Aug.2024, Second Prize in 17th CISCN Security Project Contest [certificate] *National*
- Jul.2024, Top 8 in Galaxy Generative AI Safety Contest Attack Track *National*
- Feb.2024, S Prize in 2024 Mathematical Contest In Modeling (MCM) [paper] *National*
- Nov.2023, Second Prize in 11th Digital Media Technology and Creativity Contest [certificate] *National*
- Nov.2023, Third Prize in 8th National Cryptography Contest [certificate][list][news] *National*
- Sep.2023, Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling [cert.] *Provincial*
- Sep.2023, Merit-based Scholarships For Outstanding Students *School*
- Aug.2023, Third Prize in 16th CISCN Security Project Contest [cert.][report][slide][video] *National*
- Jun.2023, Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest [certificate] *Regional*
- Apr.2023, First Prize in 13th MathorCup Mathematical Modeling Challenge [cert.][paper] *National*
- Apr.2023, First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate] *Provincial*
- Feb.2023, S Prize in 2023 Mathematical Contest In Modeling [certificate][paper] *National*
- Jun.2022, Winning Prize in 15th CISCN Security Project Contest [certificate][report][slide] *National*
- Sep.2022, Merit-based Scholarships For Outstanding Students *School*
- Jun.2022, Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate] *Regional*
- Jun.2021, Third prize in 12th Lanqiao Software Development and Algorithm Contest [cert.] *National*
- Sep.2021, Merit-based Scholarships For Outstanding Students *School*
- Apr.2021, First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate] *Provincial*

Open Source Activities

- NVIDIA/garak (Generative AI Red-teaming Tool Kit. Metasploit in the field of LLM Security. **3.0K+ Stars**)

  (i) Robustness Enhancement: Python interpreter version bug fix#296; Replicate LLM generator exception error fix#401; Add exception handling logic to the Detector base class#566;
  (ii) New Feature: Multi-modal Jailbreaking Attack on LLaVA #587 (Make it the **world's first** tool with multimodal red teaming capabilities.); Prompt Architecture Enhancement for Better Multi-modal Red Teaming (#658); Introducing new probe SciSafeEval (#948); Network Proxy Feature For Generator (garak:proxy_dev) (v0.9.0.9, v0.9.0.11, v0.9.0.13)

Academic Services

- Reliability Engineering & System Safety, Elsevier *(JCR-Q1, IF=9.4, Top)* — Reviewer
- The 31st International Conference on Computational Linguistics (COLING 2025) — Reviewer
- The 6th AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI 2025) — Reviewer
- 2025 International Joint Conference on Neural Networks (IJCNN 2025) — Reviewer
- The 3rd Deployable AI Workshop colocated at AAAI-2025 (DAI 2025) — Reviewer
- 2025 IEEE International Symposium on Biomedical Imaging (ISBI 2025) — Reviewer
- MLCommon/AI Risk & Reliability (AIRR) Working Group — Member/Individual Contributor
- Special Interest Group in Modern Interdisciplinary Research (SIGMIR) — Founder

李天昊

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教育学位

杜克大学 **2024年8月 - 2026年5月 (预计)**

理学硕士研究生 - 医学物理

研究方向: 可信人工智能；大语言模型；医学图像处理。（首年base昆山杜克）

北方工业大学 **2020年9月 - 2024年6月**

工学学士 - 信息安全 绩点: *3.89/4* 成绩: *90.88/100* 成绩单

核心课程: 应用密码学, 数据安全, 数字图像处理, 计算机组成原理, 操作系统, 逆向工程, 数字取证, 软件安全, 网络攻防

荣誉奖项: 优秀毕业论文 (排名1/101, Top 0.99%), 一等优秀学生奖学金 (排名1/329, Top 0.30%)

学术界研究经历

威斯康星大学麦迪逊分校 **2024年10月 – 至今**

研究实习 @ 安全基础模型系统实验室（*SaFoLab*） 美国，威斯康星

- **职责**: 带领科研团队推进AgentFuzzer项目，提出统一的智能体模式描述符AgentSchemaDescriptor，基于此提出首个针对大模型智能体的系统性模糊测试框架，解决了对于智能体安全对齐测试泛化性难以提升的挑战。(将投稿至ICML'25)
- **导师**: Chaowei Xiao教授

浙江大学 **2024年8月 – 2024年10月**

访问研究员 @ 知识引擎实验室 中国，浙江

- **职责**: 带领15人科研团队推进SciSafeEval项目，提出首个面向大模型科学任务安全性的基准评测，在该基准评测下的实验结果指出：包括GPT-4o/o1-preview、LLaMa3.2、QWen2.5等在内的SOTA大语言模型(LLMs)存在生物、化学、医学、物理等科学领域的技术滥用安全风险，同时给出了基于思维链(CoT)技术与小样本学习(Few-shot)技术的有效风险缓解措施。
- **产出**: arXiv:2410.03769 (ICLR'25 Sub) | Huggingface:Tianhao0x01/SciSafeEval | Github:DavidLee528/SciSafeEval
- **导师**: 张强教授，陈华钧教授

高精尖创新研究院 **2022年7月 – 2023年7月**

研究助理 @ 虚拟与增强现实技术创新中心 中国，北京

- **职责**: 带领了一个8人研发团队，开发了基于步态的情报、监视和侦察（ISR）系统。实现了步态识别SaaS原型系统，并提出了包括身份认证、跟踪和近源渗透预警在内的安全特性。负责技术架构设计、技术报告撰写以及研究论文的撰写工作。
- **产出**: 成功申请10万人民币研发经费，发表7篇SCI/EI论文，赢得4项国家级专业比赛，3次口头报告与1篇海报汇报。
- **导师**: 王立军院士

工业界研究经历

天融信 **2024年3月 – 2024年9月**

安全研究员 @ 创新技术研究部 中国，北京

- **职责**: **(i)** 支撑机器学习模型安全评估项目原型系统落地，涉及到对于对抗攻击、投毒攻击、等算法的调试以及原型系统后端评估模块的开发; **(ii)** 支撑网络钓鱼电子邮件检测项目，监督数据集的收集和改进工作。成功应用大型语言模型 (LLM) 和监督微调 (SFT) 开发出强大的检测系统。
- **领导**: 创新研究总经理王玮博士

绿盟科技 **2023年9月 – 2024年3月**

安全研究员 @ 创新研究院天枢实验室 中国，北京

- **职责**: **(i)** 设计绿盟大模型安全评估系统(NSFOCUS LSAS)核心算法与原型系统; **(ii)** 通过进攻性测试，即"红队"测试，提高绿盟风云卫大模型(NSFGPT)和其他私有LLM的安全和隐私性能; **(iii)** 进行文献综述，包括可信人工智能的研究论文和Gartner咨询报告，丰富威胁情报数据库; **(iv)** 撰写LLM安全技术白皮书。
- **影响**: 发布了绿盟大模型安全评估系统(LSAS)，绿盟风云卫大模型NSFGPT以及安全行业大模型SecLLM技术白皮书
- **领导**: 主管安全研究员王星凯博士, 首席创新官&创新研究总经理刘文懋博士

部分文章发表

- Li, T., Lu, J., Chu, C., Zeng, T., Zheng, Y., Li, M., Huang, H., Wu, B., Liu, Z., Ma, K., Yuan, X., Wang, X., Ding, K., Chen, H., & Zhang, Q.*. SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks. *International Conference on Learning Representations, 2025 (ICLR Submission)* [PDF]
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- Li, T., Zheng, Y., Zhang, H., Ma, W., & Li, Y.*. Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. *Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, 2023 (BDIoT)* [PDF][Oral][Slide]

部分荣誉奖项

- (2024年08月) 第十七届全国大学生信息安全竞赛作品赛二等奖 [证书] 国家级
- (2024年07月) 星河生成式人工智能安全大赛攻击赛道第8名（共89队） 国家级
- (2024年02月) 2024美国大学生数学建模竞赛(MCM)S奖 [论文] 国家级
- (2023年11月) 第十一届全国大学生数字媒体科技作品及创意竞赛二等奖 [证书] 国家级
- (2023年11月) 第八届全国密码技术竞赛三等奖 [证书][名单][新闻] 国家级
- (2023年09月) 全国大学生数学建模竞赛北京赛区甲组二等奖 [证书] 省部级
- (2023年09月) 优秀学生、学习、创新实践、体育、思想品德奖学金 学校级
- (2023年08月) 第十六届全国大学生信息安全竞赛作品赛三等奖 [证书][报告][幻灯片][演示][新闻] 国家级
- (2023年06月) 第十六届全国大学生信息安全竞赛实践赛(AWDP)三等奖 [证书] 区域级
- (2023年04月) 第十三届MathorCup数学建模挑战赛一等奖 [证书][论文] 国家级
- (2023年04月) 第十四届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书] 省部级
- (2023年02月) 2023美国大学生数学建模竞赛成功参与奖 [证书][论文] 国家级
- (2022年09月) 优秀学生、学习、创新实践、体育、思想品德奖学金 学校级
- (2022年06月) 第十六届全国大学生信息安全竞赛作品赛优胜奖 [证书][报告][幻灯片] 国家级
- (2022年06月) 第十五届全国大学生信息安全竞赛实践赛(CTF)三等奖 [证书] 区域级
- (2021年09月) 学习、创新实践奖学金 学校级
- (2021年06月) 第十二届蓝桥杯全国软件和信息技术专业人才大赛三等奖 [证书] 国家级
- (2021年04月) 第十二届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书] 省部级

开源活动

- NVIDIA/garak (生成式人工智能红队工具，大模型安全领域的Metasploit，**3.0K+ Stars**)

  (i) 鲁棒性提升: Python解释器版本错误修复#296; Replicate大模型生成器异常修复#401; Detector检测器基类增加异常处理逻辑的增加#566;
  (ii) 新特性引入: 针对LLaVA的多模态越狱攻击#587 (使其成为**世界上首个**具备多模态红队能力的工具); 多模态红队提示词探针架构升级 (#658); 全新探针SciSafeEval的引入 (#948); 生成器网络代理特性 (garak:proxy_dev). (v0.9.0.9, v0.9.0.11, v0.9.0.13)

学术服务

- Reliability Engineering & System Safety, Elsevier *(JCR-Q1,* 中科院一区*, IF=9.4, Top)* 审稿人
- The 31st International Conference on Computational Linguistics (COLING 2025) 审稿人
- The 6th AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI 2025) 审稿人
- 2025 International Joint Conference on Neural Networks (IJCNN 2025) 审稿人
- The 3rd Deployable AI Workshop colocated at AAAI-2025 (DAI 2025) 审稿人
- 2025 IEEE International Symposium on Biomedical Imaging (ISBI 2025) 审稿人
- MLCommon/人工智能风险与可靠性(AIRR)工作组 成员/独立贡献者
- Special Interest Group in Modern Interdisciplinary Research (SIGMIR) 创始人